Analysis of Data from Multiclinic Trials

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ABSTRACT Because the clinics in a multiclinic randomized clinical trial represent neither fixed stratification effects nor random classificatory effects, the appropriate analysis of data from such a trial has been the subject of controversy and debate. The following are some of the elements of controversy that are discussed and for which some bases for resolution are proposed. Is it ever valid to ignore the effects of clinics in the analysis? Is it ever valid to drop clinics from the analysis? Is a multiclinic clinical trial similar in structure or not to a single-clinic clinical trial in which patients have been stratified on a classificatory factor? Assuming that clinics will be taken account of in the analysis, should it be the weighted or the unweighted average of within-clinic treatment differences that is to be taken as the best estimate of the overall difference between the treatments? How should the data be analyzed if there is evidence of treatment-by-clinic interaction?

KEY WORDS: *multiclinic clinical trial, pooling of data, treatment-by-clinic interaction*

INTRODUCTION

A multiclinic trial is one conducted in a number of participating clinics simultaneously. What makes such a trial a single study rather than a series of separate, unrelated studies is the adherence to a common protocol, in theory if not in actuality. Multiclinic trials are at least as popular today as they were in the 1940s and 1950s when Bradford Hill designed the classical controlled multiclinic studies of antihistamines, cortisone, and streptomycin [1]. The major reason for a multiclinic rather than a single-clinic trial is now, as it was then, the need to enroll sufficiently many patients so that the study has adequate power to detect a difference between the treatments being compared. The ability to generalize the study's results to more than one kind of patient and more than one kind of treatment facility is a second important reason.

Given the relatively long history of multiclinic trials, and given the relatively large literature on the planning and execution of such studies [2–9], there is a striking paucity of articles and chapters in books on the analysis of the

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resulting data. This dearth in the literature is especially striking because uncertainty and controversy exist concerning most aspects of analysis. Several problematic areas of data analysis are identified in the article, and criteria are offered for resolving some of the disagreements and ambiguities that exist. Attention is restricted throughout the article to a measured response variable that is assumed to be normally distributed.

It is important to specify at the outset what is probably the key assumption underlying the analysis, criticisms, and recommendations that follow: clinics will vary one from another in the overall mean levels of response of their patients. One set of reasons includes the enrollment of demographically as well as medically different kinds of patients, and the existence of different treatment milieux. A second set of reasons includes different types and levels of departure from the study's protocol, and differing criteria for evaluating response to treatment. Both sets of factors are likely to produce clinic-to-clinic variation in the mean level of response. The second set of factors is, as well, likely to produce treatment-by-clinic interaction (i.e., nonconstancy across the clinics of the differences among the treatments' means), whereas the first set might not.

"POOLING" THE DATA

What may be the most important controversy concerns the meaning and the strategy of "pooling." Some notation will help here. Assume for simplicity that two treatments are being compared in each of C clinics and that the underlying variances are equal for all clinics and for both treatments. Further, assume for now that there is no treatment-by-clinic interaction. Two randomization procedures are considered, one employing separate and independent randomization schedules for the several clinics and the second ignoring the clinics in the random assignment of patients to treatment groups.

The results within a typical clinic, say clinic c, may be summarized as follows.

Treatment	Sample Size	Mean	Standard Deviation
1	n_{c1}	$\overline{X}_{c1}$	s_{c1}
2	n_{c2}	$\overline{X}_{c2}$	s_{c2}

To some, "pooling" means "averaging within-clinic differences" and is thus used in the same sense as "pooling variances." Specifically, pooling in this sense means taking

$$\overline{D}_W = \frac{\Sigma W_c(\overline{X}_{c1} - \overline{X}_{c2})}{\Sigma W_c} = \frac{\Sigma W_c D_c}{\Sigma W_c} \tag{1}$$

as the estimator of $\mu_1 - \mu_2$, the assumed common difference between the treatments' means, where $D_c = \overline{X}_{c1} - \overline{X}_{c2}$ and the W_cs are a set of positive weights; and taking

$$\text{se}(\overline{D}_W) = \frac{s_p}{\Sigma W_c}\sqrt{\Sigma W_c^2 \frac{n_{c1} + n_{c2}}{n_{c1}n_{c2}}} \tag{2}$$

as its estimated standard error, where $s_p^2 = \Sigma\Sigma(n_{ci} - 1)s_{ci}^2/\Sigma\Sigma(n_{ci} - 1)$, the pooled variance with, say, $\nu = \Sigma\Sigma(n_{ci} - 1)$ degrees of freedom. The weighting systems most frequently employed either take W_c to be a constant for all c, or take W_c to be a function of n_{c1} and n_{c2}.

To others, however, "pool the data" is a euphemism for "throw together all the responses to a treatment, ignoring the clinics." Specifically, pooling in this second sense means taking

$$\overline{D} = \frac{\Sigma n_{c1}\overline{X}_{c1}}{n_{.1}} - \frac{\Sigma n_{c2}\overline{X}_{c2}}{n_{.2}} \tag{3}$$

as the estimator of the difference between the treatments' means, where $n_{.i} = \Sigma n_{ci}$ for $i = 1$ and 2, and taking

$$\text{se}(\overline{D}) = s^* \sqrt{\frac{n_{.1} + n_{.2}}{n_{.1}n_{.2}}} \tag{4}$$

as its estimated standard error, where $(s^*)^2 = [(n_{.1} - 1)s_1^2 + (n_{.2} - 1)s_2^2]/(n_{.1} + n_{.2} - 2)$, with s_i^2 being the variance of the $n_{.i}$ responses to treatment i for $i = 1$ and 2.

Under either of the randomization procedures considered here, randomization with regard or without regard to the clinics, both $\overline{D}_W$ and $\overline{D}$ are unbiased estimators of $\mu_1 - \mu_2$. The sampling variation of $\overline{D}$ exceeds that of $\overline{D}_W$ for appropriate choices of weights [10], however, and the standard deviation s^* that is used in analyses based on $\overline{D}$ will tend to be greater than the standard deviation s_p that is appropriate when $\overline{D}_W$ is the estimator (s^*will tend to be inflated relative to s_p because it will be affected by the differences among the clinics' means that are assumed always to exist).

Pooling in the first sense is the correct method of analysis whenever the random assignment of treatments to patients is carried out separately and independently within the clinics, for the analysis will then properly have been dictated by the design. Pooling in the second sense is theoretically valid only when the study design calls for randomizing without regard to clinic, so that it is theoretically possible for most or even all of a clinic's patients to be assigned the same treatment. Such randomization schemes are sometimes employed [3,11,12], mainly in cancer trials in which stratification on stage of disease is considered to be more important than control for the effects of clinics. For trials in diseases other than cancer, randomizing without regard to clinic seems rare. Even in such cases, it would be advisable to summarize the data first within clinics and then to average the treatment differences across the clinics. The strategy would therefore be analogous to post-stratification [13–15], that is, the control only in the analysis, not in the design, for a classificatory factor.

In summary, pooling in the sense of averaging within-clinic differences is almost always justified, and pooling in the sense of throwing together all the data is only rarely justified. When the design calls for separate and independent randomizations within the clinics, pooling in the latter sense is inappropriate on theoretical grounds. When the design calls for randomization without regard to clinics, it may be inefficient. The practice of pooling the data in the sense of lumping them together should generally be avoided, and

the word "pooling," given its two contradictory meanings, should also be avoided or, if not, defined whenever used.

DROPPING CLINICS FROM ANALYSIS

The practice of dropping entire clinics rrom analysis is not widespread, but it does occur. The protocol of one multiclinic study, for example, stipulated that "each center must have a minimum of 30 . . . enrollees to be included in a pooled analysis." A recent article on multiclinic trials suggested that "an entire institution can be dropped from the study if the percentage of non-valid patients . . . becomes too large or if the entry rate is not high enough" [16, p. 854]. In a personal communication, the senior author explained that the dropping referred both to withdrawal of permission to enroll future patients and to deleting from analysis the data collected on the patients already enrolled.

If enhanced prestige is a consequence of a clinic's having participated in a multiclinic trial, it is understandable that the threat of being dropped may provide an impetus to the enrollment of adequate numbers of patients, and to the enrollment of patients who satisfy the admission criteria enumerated in the protocol. Actually to carry out such a threat is to establish a precedent that may lead to bias. One may imagine a study whose sponsors are not totally blinded to treatment. Noting that the results are going in the "wrong" direction in a certain clinic, they may decide to drop that clinic with respect to recruiting future patients and thereby succeed in having it dropped from analysis because it failed to enroll the prespecified minimum number of patients.

The only valid reason I can think of for dropping a clinic from analysis is that all of its patients had been assigned the same treatment. Otherwise, the aphorism that describes the consensus view concerning which patients to analyze in a clinical trial, "if randomized, then analyzed," should apply to clinics as well.

TAKING THE CLINIC EFFECTS TO BE RANDOM

A question exists as to just how the clinics are to be taken account of in the analysis, as fixed or as random effects. The most compelling presentation of a model that takes the clinic effects to be random is in a paper by Chakravorti and Grizzle [17]. The resulting statistical model for the obtained data is a mixed model, with the effects of treatments being fixed and the effects of clinics being random. Even though such a model is never strictly valid, inasmuch as the participating clinics are never sampled randomly or even haphazardly from either an infinite or a finite population of clinics [18], it might be tenable when there are relatively many clinics and relatively few patients in each [19]. Under this model, linear combinations of the pooled variance, s_p^2, and the interaction mean square,

$$\text{IMS} = \frac{\sum_{c=1}^{C} W_c^* (D_c - \overline{D}_{W*})^2}{C - 1} \tag{5}$$

with $C - 1$ degrees of freedom, must be used in making inferences about the difference between the treatments' means. In equation (5),

$$W_c^* = \frac{n_{c1}n_{c2}}{n_{c1} + n_{c2}} \tag{6}$$

and

$$\overline{D}_{W^*} = \frac{\Sigma W_c^* D_c}{\Sigma W_c^*}. \tag{7}$$

Except under special circumstances, the analytical methods are complicated and the test procedures only approximate [17,18].

Generally, the clinics have more of the earmarks of a fixed factor than of a random factor, and thus analyses assuming a mixed model would be unnecessarily complicated. In fact, clinics are usually deliberately selected because of a reputation for having conducted clinical trials, because of an expectation of success in enrolling the requisite numbers of patients and in adhering to the study protocol, because of the unique kinds of patients they treat, and so on. Were the study to begin at a different time, most if not all of the participating clinics would be the same.

The consensus in the field seems to be that, except in certain circumstances [20], clinics are more appropriately modeled as representing levels of a fixed factor than levels of a random factor. The designation of a factor as fixed is not sufficient, however, to indicate how the data are to be analyzed and the results interpreted. Both classificatory and bona fide experimental factors are considered to represent levels of a fixed factor, after all, but how their effects are defined, estimated, and analyzed depends on whether assignment to one level or another is under the investigator's control [21, pp. 92–93]. Clinics obviously represent a classificatory, not an experimental factor.

ARE CLINICS LIKE STRATA?

Even identifying a factor as classificatory rather than experimental is insufficient for specifying how to conceive of and analyze its effects. Consider Greenberg's suggestion that clinics be considered similarly to such strata as stages of disease [22]. The analogy is an imperfect one, and it is especially weak when there is treatment-by-clinic interaction.

In the brief theoretical development given earlier, suppose that all the measurements were obtained on patients randomly sampled from a single clinic, with the patients having been stratified into C stages of disease. The classical method of analysis [23, pp. 264–265; 24, pp. 150–153] calls for the weighting factor W_c in equation (1) to be set equal to W_c^* in equation (6) for $c = 1, \ldots, C$. The resulting test is based on the magnitude of the statistic

$$t = \frac{\Sigma W_c^* D_c}{s_p \sqrt{\Sigma W_c^*}} \tag{8}$$

with $\nu = \Sigma\Sigma(n_{ci} - 1)$ degrees of freedom.

When there is no treatment-by-stratum interaction, this weighting system is optimal in that the resulting weighted average in equation (1) is the min-

imum variance unbiased estimator of the common within-stratum difference between treatment means. When there is treatment-by-stratum interaction, this weighting system, although no longer optimal in any rigorous sense, is at least valid and defensible in that $W_c^*/\Sigma W_c^*$, is a consistent estimator of the proportion of patients in the study population who are from stratum c [24, p. 162], and thus $\overline{D}_{W*}$ in equation (7) is a consistent estimator of the difference between the two treatments' means in that population.

Consider, now, the apparently analogous multiclinic study, with C representing, as earlier, the number of clinics. When no treatment-by-clinic interaction exists, the weights given in equation (6) are optimal in the same sense as in a stratified study. When, on the other hand, such interaction exists, those weights can no longer be interpreted as they were in the case of a stratified study. In no easily stated sense is there a definable population of patients that was sampled from. A clinic's W_c^* may be informative about how many eligible patients sought treatment there, but it may be just as informative about how vigorous clinic c was in enrolling and retaining patients. The earlier informativeness of W_c^* about the relative size of stratum c is thus no longer pertinent.

THE DETECTION OF INTERACTION

The most challenging questions in the analysis of the data from a multiclinic trial are how to carry out the analysis when there is treatment-by-clinic interaction, and, prior to that, how to ascertain whether such interaction exists.

The U.S. Food and Drug Administration issued draft guidelines in June 1985 for the format and content of a new drug application's statistics section [25]. The importance it places on treatment-by-clinic interaction is seen in such recommendations as "The documentation of multi-investigator studies should include at a minimum an analysis of variance table with terms for investigators, treatments, (and) their interaction" (p. 19), "(present plots or graphs so as to) convey . . . information about consistency of drug effects across investigators" (p. 20), and "(as) rationale . . . for combining results across investigators, . . . suitable consistency of results across clinics should be documented" (p. 22). The test statistic for interaction in the analysis of variance is the ratio $F = \text{IMS}/s_p^2$ with $C - 1$ and ν degrees of freedom, with IMS as defined in equation (5). Unfortunately the draft guidelines provide neither warning nor reassurance as to the significance level at which the test should be performed, nor do they help determine the degree of inconsistency of results across clinics that would make the combination of results unsuitable.

Some statistical reviewers at the FDA have, allegedly, required applicants to consider the problem of interaction—and, perhaps, to take account of it in the analysis—if the F ratio for interaction is significant at the 0.10 level. Even though the chances are one in ten that there will be undue concern about interaction and perhaps even an inefficient analysis when interaction is not really there, the relatively high power that such a criterion provides when in fact there is interaction is reassuring to those who believe that clinical or demographic differences between the clinics' patients make interaction possible, and that loose controls and little or no monitoring of procedures at the

individual clinics make interaction inevitable [10]. I recommend basing the decision whether or not there is interaction on the test of significance described above, with a significance level of 0.10.

(Interactions due to factors inherent in the clinics—their patients or their treatment milieux—are impossible to eliminate, but the gratuitous interactions due to idiosyncratic interpretations or violations of the study's protocol may be eliminated or at least minimized by the adoption of certain procedures. Regular visits to or telephone contacts with each of the clinics, the consequent early detection of protocol violations, and thus the quick imposition of remedial controls are superior to leaving things out of control and thus running the risk of large interactions that may make the study's results uninterpretable. How to carry out the necessary monitoring, and what kinds of problems to look out for, are both known [2,4–6,9].)

ANALYSIS IN THE PRESENCE OF INTERACTION

Despite experimental prevention's theoretical superiority to statistical cure, it may be too expensive to apply or, if applied, ineffective in preventing interaction. Assuming that treatment-by-clinic interaction has been found to be significant, how should the data be analyzed in order to help answer the study's single primary question, Is there a difference between the two treatments' means? As was suggested by Yates over 50 years ago [26], the mean difference that makes sense when there is interaction is $\Sigma(\mu_{c1} - \mu_{c2})/C$, where μ_{ci} is the underlying mean response to treatment i within clinic c ($i = 1,2$; $c = 1, \ldots, C$). The hypothesis being tested is that the simple average of the differences $\mu_{c1} - \mu_{c2}$, with no clinic receiving greater or lesser weight than another, is zero. The pertinent test statistic when interaction has been found to exist is

$$t = \frac{\Sigma D_c}{s_p \sqrt{\frac{1}{W_c^*}}} \tag{9}$$

with ν degrees of freedom, which is equal to the ratio of the mean in equation (1) to the standard error in equation (2) for a weighting system that takes $W_c = 1$ for each value of c. Giving each clinic's estimated mean difference the same weight is easily explained to one's clinical colleagues when the C sample sizes are of comparable magnitudes but may be extremely difficult to defend when some of the sample sizes $n_{c1} + n_{c2}$ are small (2 or 3, for example) and others large (10 or more times greater). Nevertheless, the test based on the statistic in equation (9) is theoretically correct when interaction exists, regardless of the sample sizes.

As an alternative to performing this test when the F ratio for interaction is significant, Overall [27, pp. 69–70] suggested the following procedure: (1) identify and then eliminate from the analysis the clinic that contributes most to the mean square for interaction; (2) check whether the elimination of that clinic succeeds in reducing the magnitude of interaction to nonsignificance; and, if so, (3) test for the significance of the mean difference between treatments by applying the test statistic in equation (8) to the data from the clinics

that remain. If the interaction remains statistically significant, the process would be continued. The Type I error rate for Overall's suggested procedure would be improved by imposing control for multiple comparison artifacts, perhaps by comparing the absolute value of the test statistic in equation (8) to the Scheffé criterion $\sqrt{CF_{\alpha,C,\nu}}$. The fact that data had to be discarded in order for a consistent, statistically significant difference to be obtained, however, may damage the credibility of the final result.

Yet other procedures have doubtless been proposed or applied when interaction is present, but the relevant literature in which they are presented, analyzed, and criticized does not seem to exist. Perhaps this review article will provide an impetus for publication, debate, and, ultimately, consensus on how best to analyze the data from a multiclinic trial when treatment-by-clinic interaction exists.

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